



Amazon Prime Video Content Analysis

Exploratory Data Analysis on **9,871 titles** and **124,000+ credits** — uncovering patterns in content, quality, and global production.

Project Objectives



Genre Dominance

Identify which genres and categories lead the platform



Release Trends

Analyze content distribution across decades



Top Performers

Identify highest-rated and most popular titles



Global Hubs

Examine geographical production and genre dominance

Dataset Overview

9,871

Title Records

15 variables: type, genre, scores, runtime, country

124K+

Credit Entries

Actors, directors, and roles per title

Key Variables

- Title, Type (Movie / Show)
- Release year, runtime, genres
- Production countries
- IMDb & TMDB scores

Data Cleaning

01

Duplicates Removed

e.g. *The Gunman From Bodie, Dynamite*

02

Missing Values Handled

age_certification 65% missing · seasons 86% missing

03

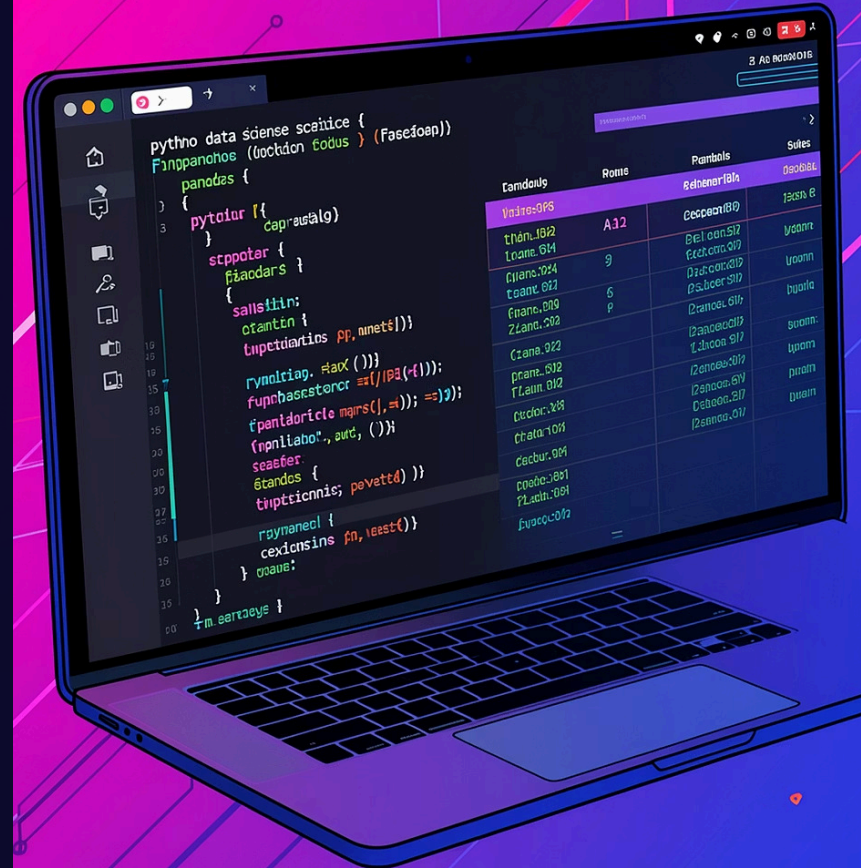
Type Transformation

Numerical & categorical columns verified

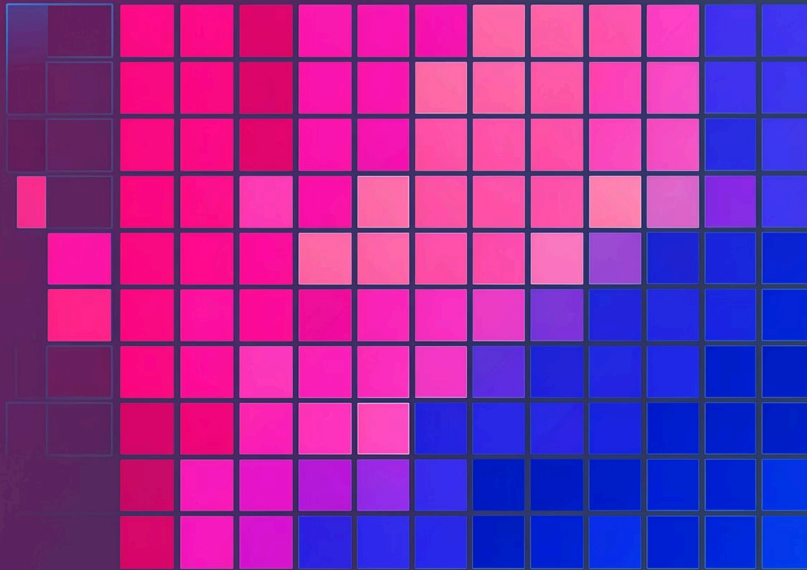
04

Outlier Treatment

Runtime & release year anomalies resolved



Exploratory Data Analysis



Three-Layer Approach

Univariate

Release year distribution — negatively skewed toward recent content

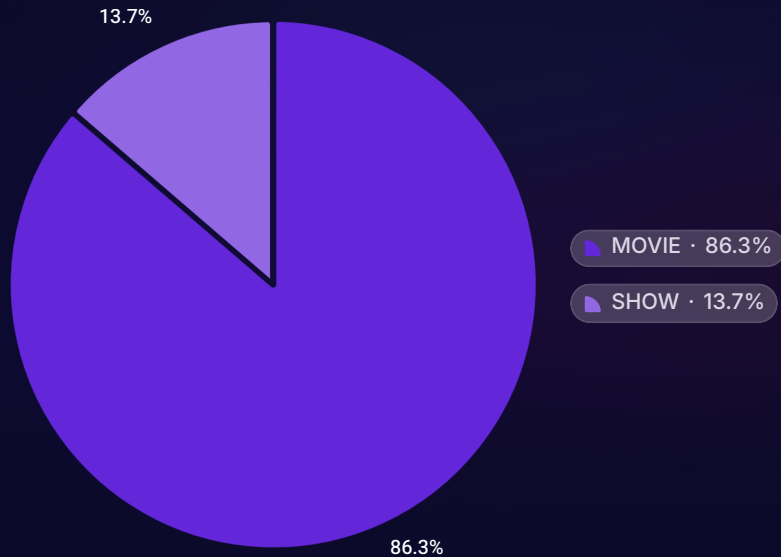
Bivariate

IMDb scores compared across genres and content types

Correlation

Heatmaps linking popularity, runtime, and ratings

Movie vs. Show Split



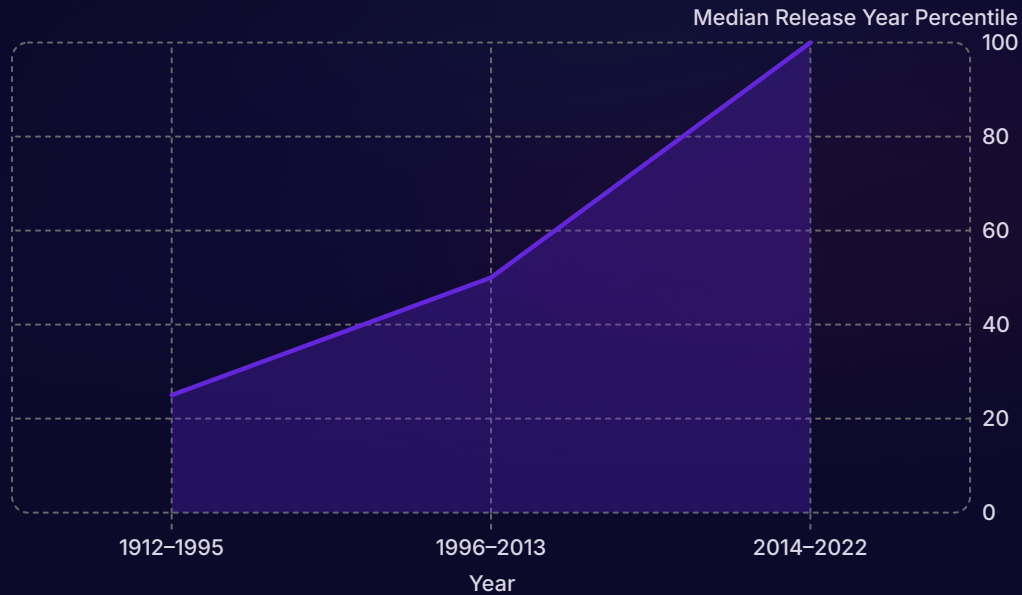
86.3%

Movies dominate the catalog

Only **13.7%** of titles are TV Shows — a significant imbalance with implications for user retention.

i Investing in TV Shows could boost long-term engagement.

The Streaming Boom

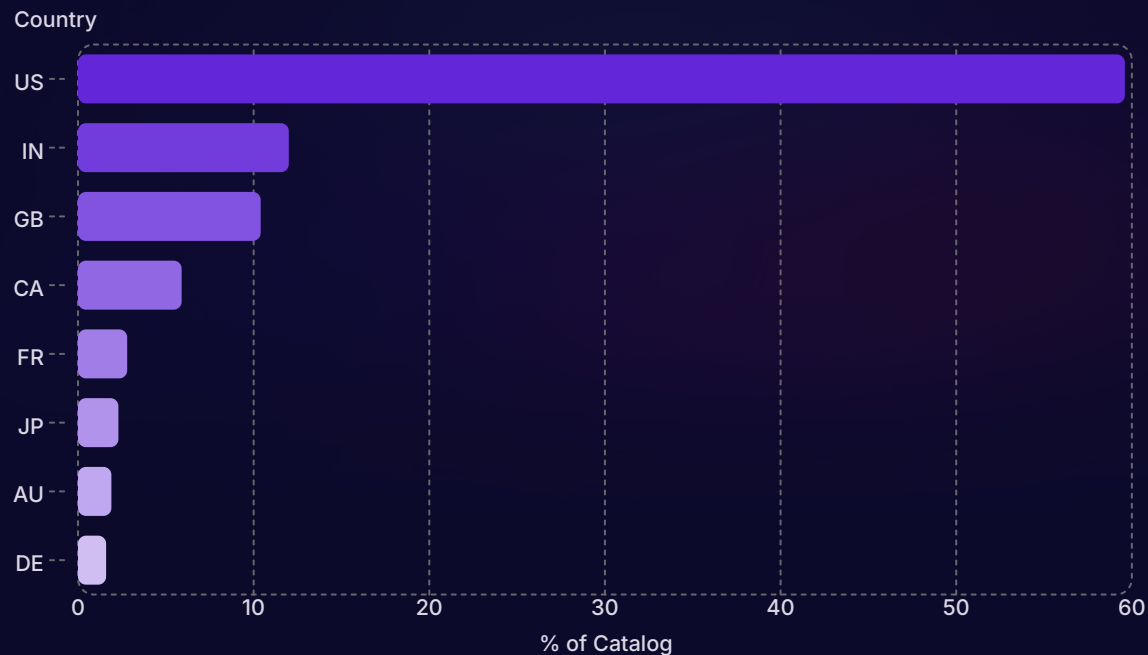


Nearly 50% of the library released after 2014

- Median release year: **2014**
- Min: 1912 · Max: 2022
- 75th percentile: **2018**

Aggressive recent expansion reflects the streaming wars era.

Global Production Hubs



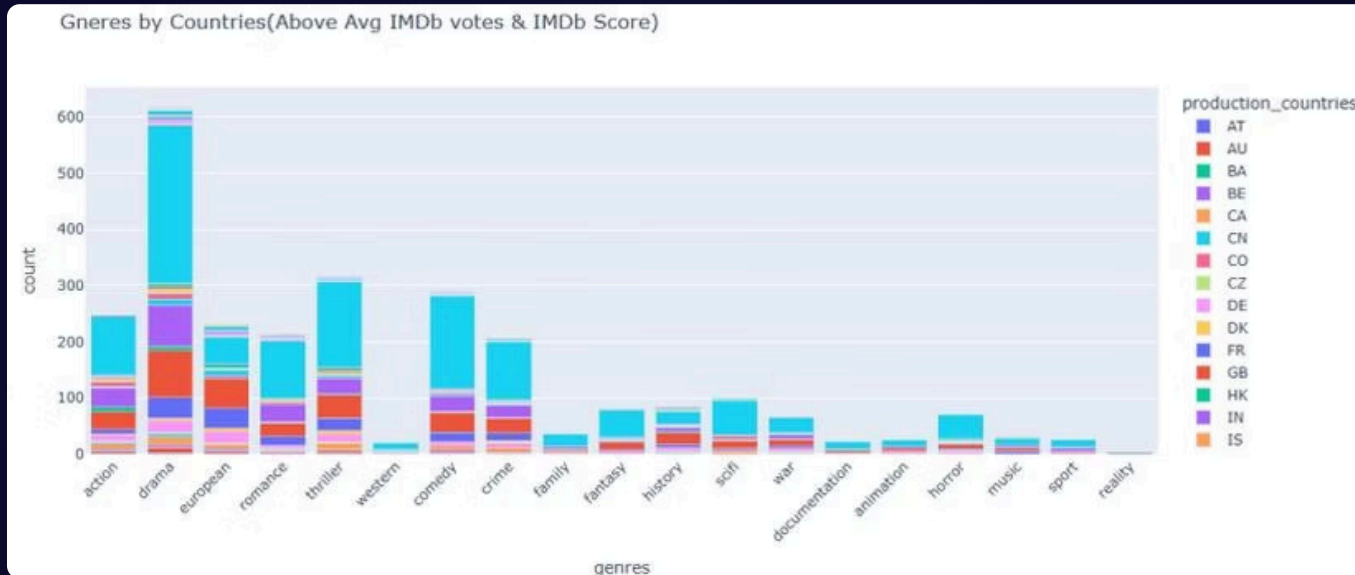
US Dominates

59.6% of all content originates from the United States.

India is a rising force at **12%** — a significant regional market.

Non-US content acquisition remains a key growth opportunity.

Genre Performance & Runtime Insights



Top Genres

Runtime Sweet Spot

Most titles: **80–120 min**. Longer films (130 min) score highest on IMDb (~6.8). Weak link between TMDB popularity and IMDb score — trending ≠ best-rated.

Drama

Thriller

Comedy

Action

Business Recommendations

1

Non-US Acquisition

Expand international content to capture untapped audiences

2

Pre-2014 Classics

Acquire high-performing older titles for nostalgia-driven engagement

3

Invest in TV Shows

Boost user retention and long-term platform engagement

4

Runtime 80–120 min

Prioritize content in the proven feature-length sweet spot

5

Focus on Top Genres

Drama, Thriller, Comedy, and Action drive the highest performance